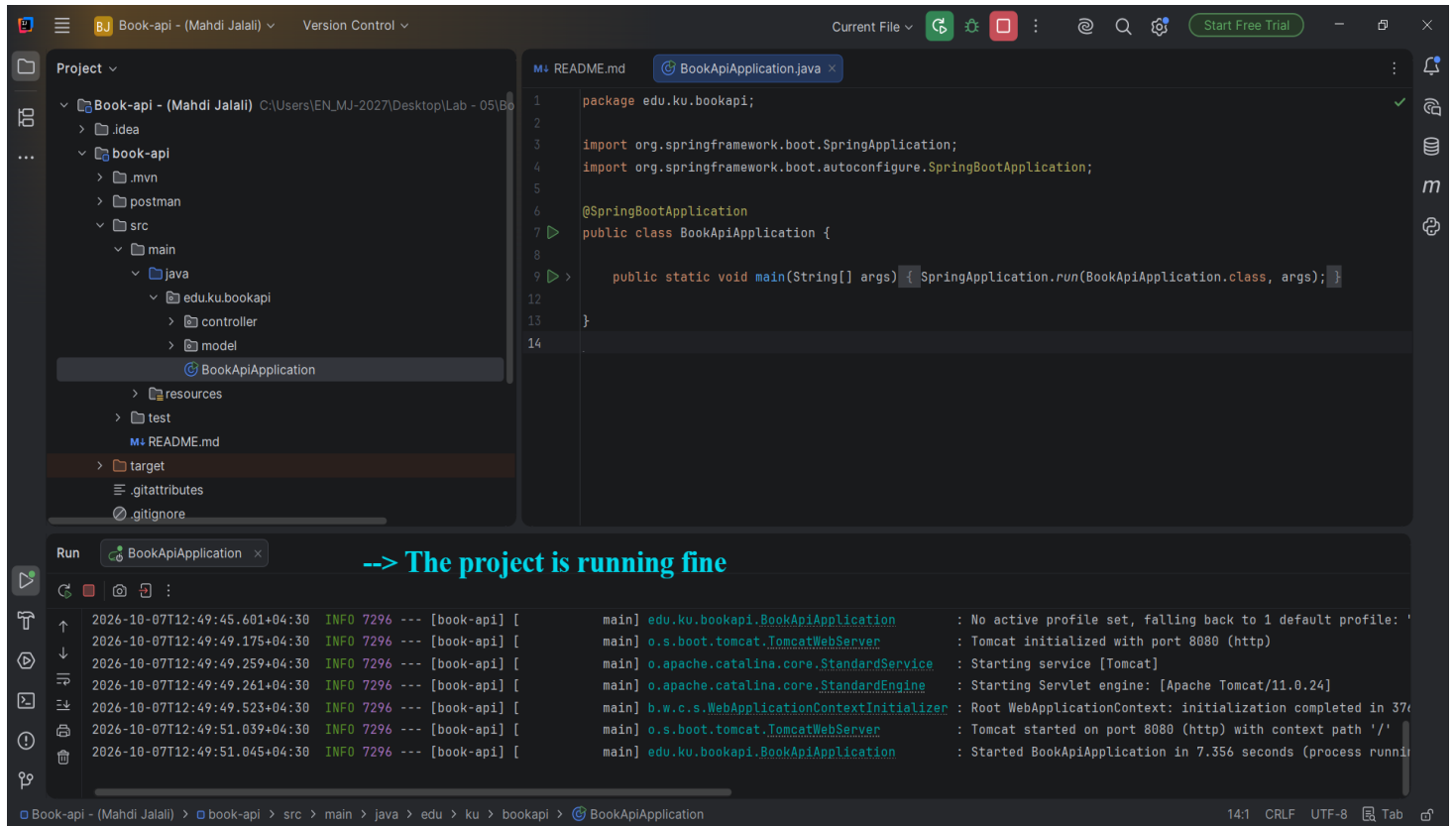


In the name of Allah

Lab-05 assignment: Connecting the front end using Axios to Back end

1. This is the back files and codes that is running fine after running we should test the files running in the browser in localhost:8888



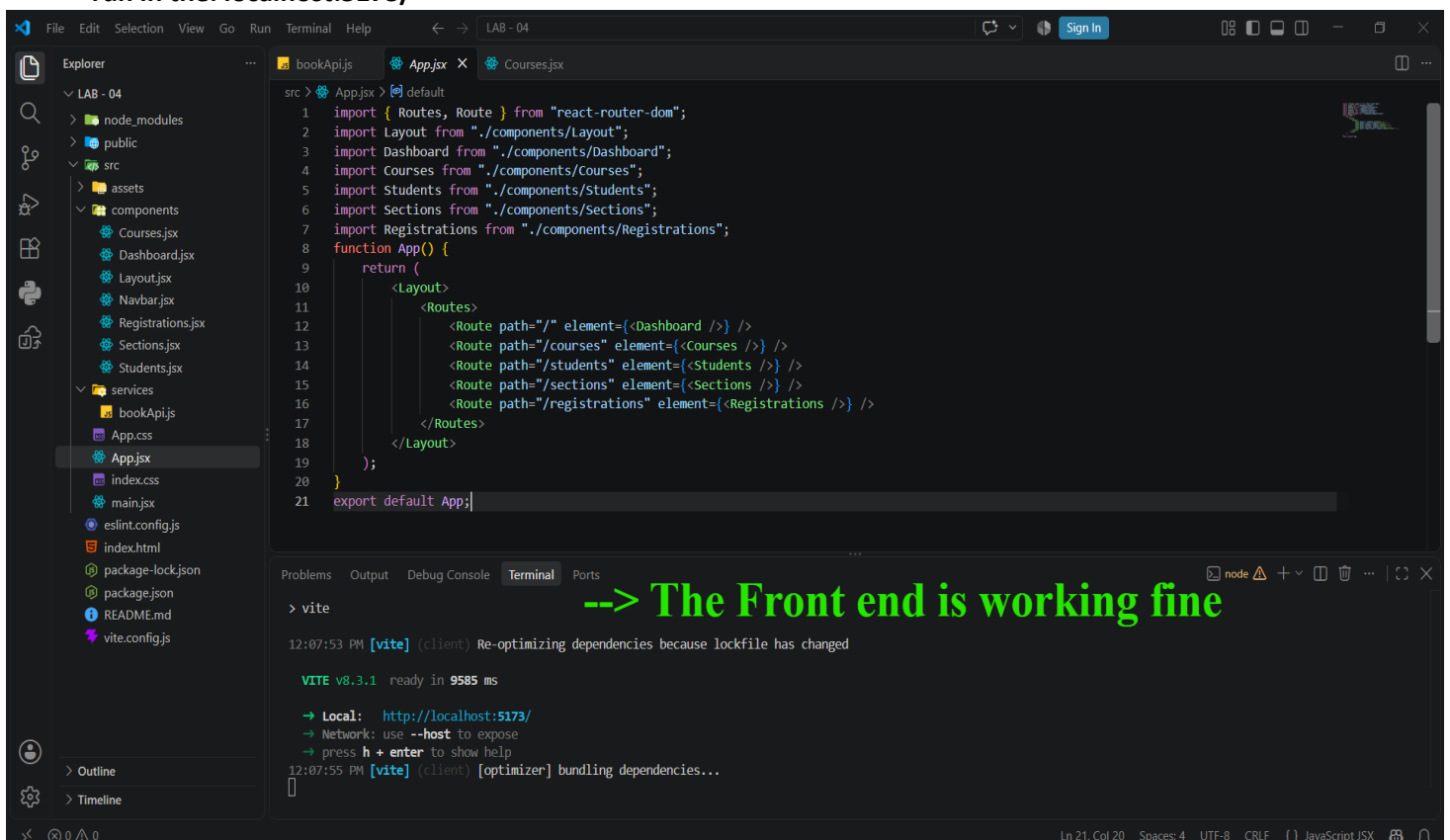
The screenshot shows an IDE with the project 'Book-api - (Mahdi Jalali)' open. The file 'BookApiApplication.java' is selected, showing the following code:

```
1 package edu.ku.bookapi;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class BookApiApplication {
8
9     public static void main(String[] args) { SpringApplication.run(BookApiApplication.class, args); }
10
11 }
12
13
14
```

The Run console shows the output of the application, indicating it is running successfully:

```
--> The project is running fine
2026-10-07T12:49:45.601+04:30 INFO 7296 --- [book-api] [main] edu.ku.bookapi.BookApiApplication : No active profile set, falling back to 1 default profile: '
2026-10-07T12:49:49.175+04:30 INFO 7296 --- [book-api] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-10-07T12:49:49.259+04:30 INFO 7296 --- [book-api] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-10-07T12:49:49.261+04:30 INFO 7296 --- [book-api] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-10-07T12:49:49.523+04:30 INFO 7296 --- [book-api] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 374
2026-10-07T12:49:51.039+04:30 INFO 7296 --- [book-api] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-10-07T12:49:51.045+04:30 INFO 7296 --- [book-api] [main] edu.ku.bookapi.BookApiApplication : Started BookApiApplication in 7.356 seconds (process running)
```

2. These are the front-end files that is working fine after running the react project, the react project will be run in the: localhost:5173/



The screenshot shows an IDE with the project 'LAB - 04' open. The file 'App.jsx' is selected, showing the following code:

```
1 import { Routes, Route } from "react-router-dom";
2 import Layout from "./components/Layout";
3 import Dashboard from "./components/Dashboard";
4 import Courses from "./components/Courses";
5 import Students from "./components/Students";
6 import Sections from "./components/Sections";
7 import Registrations from "./components/Registrations";
8 function App() {
9     return (
10         <Layout>
11             <Routes>
12                 <Route path="/" element={<Dashboard />} />
13                 <Route path="/courses" element={<Courses />} />
14                 <Route path="/students" element={<Students />} />
15                 <Route path="/sections" element={<Sections />} />
16                 <Route path="/registrations" element={<Registrations />} />
17             </Routes>
18         </Layout>
19     );
20 }
21 export default App;
```

The Terminal shows the output of the application, indicating it is running successfully:

```
--> The Front end is working fine
12:07:53 PM [vite] (client) Re-optimizing dependencies because lockfile has changed
VITE v8.3.1 ready in 9585 ms
→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
12:07:55 PM [vite] (client) [optimizer] bundling dependencies...
```

3. This is the browser that show the connectivity of front end to the back end, and we can see the course content shown to us.

CompusHub Dashboard Courses Students Sections Registrations

Book Inventory Management
Total Books Available: 5

ID	Title	Author	ISBN	Published Year	Category
1	Clean Code	Robert C. Martin	9780132350884	2008	Software Engineering
2	Effective Java	Joshua Bloch	9780134685991	2018	Java
3	Designing Data-Intensive Applications	Martin Kleppmann	9781449373320	2017	Distributed Systems
4	Spring in Action	Craig Walls	9781617297571	2022	Spring
5	Computer Networks	Andrew S. Tanenbaum	9780132126953	2010	Networking

-> These are the content of the Courses.jsx file that is shown in here

* -> I also added a loading action when the front wants to talk to the back end and get the content from backend when the front is starting

4. These are the books that is stored in the back files

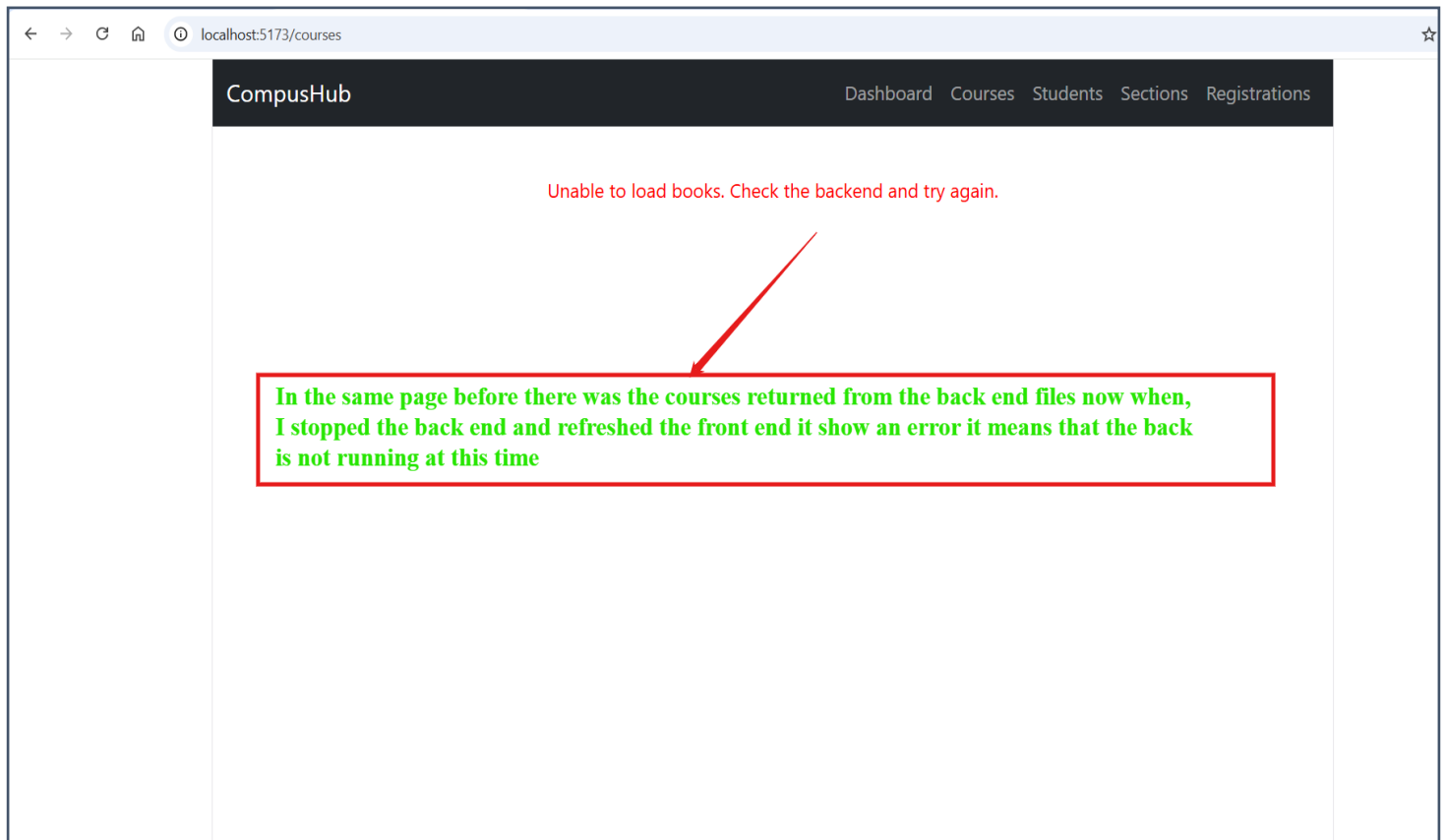
localhost:8080/api/v1/books

Pretty-print ☒

```
[
  {
    "id": 1,
    "title": "Clean Code",
    "author": "Robert C. Martin",
    "isbn": "9780132350884",
    "publishedYear": 2008,
    "category": "Software Engineering"
  },
  {
    "id": 2,
    "title": "Effective Java",
    "author": "Joshua Bloch",
    "isbn": "9780134685991",
    "publishedYear": 2018,
    "category": "Java"
  },
  {
    "id": 3,
    "title": "Designing Data-Intensive Applications",
    "author": "Martin Kleppmann",
    "isbn": "9781449373320",
    "publishedYear": 2017,
    "category": "Distributed Systems"
  },
  {
    "id": 4,
    "title": "Spring in Action",
    "author": "Craig Walls",
    "isbn": "9781617297571",
    "publishedYear": 2022,
    "category": "Spring"
  },
  {
    "id": 5,
    "title": "Computer Networks",
    "author": "Andrew S. Tanenbaum",
    "isbn": "9780132126953",
    "publishedYear": 2010,
    "category": "Networking"
  }
]
```

all the books stored in the backend files

5. Here we can check that if the front end is really connected to the back end so I stopped the backend files and this error shown to us as expected



Thank you!